A Mini Project Report on

AI-Based Autonomous Surveillance Robot for Police Applications

Submitted to the

DEPARTMENT OF ELECTRONICS AND INSTRUMENTATION ENGINEERING

In partial fulfillment of the requirement of

Bachelor of Engineering in Electronics and Instrumentation Engineering

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June 2026

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CERTIFICATE

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Declaration

We hereby declare that the dissertation ***"AI-Based Autonomous Surveillance Robot for Police Applications"*** submitted by us to the department of Electronics and Instrumentation Engineering, RIT Bengaluru-560 054 in partial fulfillment of the requirement of the course EIP67-Mini project of **Bachelor of Engineering** in **Electronics & Instrumentation Engineering** is a bona-fide record of the work carried out by us under the supervision of ***Dr. Nishi Shahnaj Haider*** .
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Acknowledgements

The gratification of happiness that accompanies any successful work would be incomplete without the mention of the people who made it possible.

We would like to express our gratitude to our guide **Dr. Nishi Shahnaj Haider**, Assistant Professor, for providing active support and guidance at every stage of our project.

We also thank **Dr. K M Vanitha**, our Mini project coordinator, for her timely guidance and help throughout the project.

We consider it a privilege to thank **Dr Shivaprakash G**, Professor and Head of the department of Electronics and Instrumentation Engineering, Ramaiah Institute of Technology, Bengaluru for providing all the facilities. We thank him for sharing the LaTeXtemplate for the Preparation of the Major Project report. We are also thankful to him for taking time to train us regarding how to use the LaTeXtemplate. We take this opportunity to thank our project guide Guide name here for timely, and valuable suggestions.

We thank **Dr. N. V. R. Naidu**, Principal and Management of Ramaiah Institute of Technology, Bengaluru for all the support to carry out the project.

We whole heartily thank the Management of Ramaiah Institute of Technology for providing us a vibrant learning ecosystem for our all round development.

We are also thankful all the faculty and staff of Electronics and Instrumentation Engineering Department and our friends for playing an important role in the completion of the Mini project.

We express sincere thanks to our parents and family for providing us with all the amenities required to complete the project and being our sole inspiration throughout the tenure of this project.

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Abstract

The rapid urbanization of modern cities has introduced complex challenges for law enforcement, particularly in the areas of traffic management and public safety. This project presents the design and implementation of an AI-Based Autonomous Surveillance Robot specifically engineered to assist police departments in routine patrolling and automated enforcement tasks. At its core, the system utilizes a specialized dual-processor hardware architecture designed to decouple high-level cognitive tasks from low-level physical actuation. A Raspberry Pi 5 serves as the Master Control Unit, providing the computational power necessary to run a YOLO (You Only Look Once) deep learning model for real-time object detection. Simultaneously, an Arduino Uno operates as the Slave Control Unit, managing a suite of sensors including IR line-trackers and an ultrasonic sensor mounted on a servo motor to facilitate autonomous navigation and obstacle avoidance.

The robot is programmed to maintain a constant patrol along a predetermined path while its vision system captures high-definition frames via a 5MP Pi Camera. The integrated AI model is trained to identify three critical environmental triggers: "No Parking" signs, potential weapons, and human falls. Upon the detection of a traffic violation, the robot initiates an Automatic License Plate Recognition (ALPR) sequence, capturing vehicle data and transmitting it to a central database for administrative action. In the event of a security threat or medical emergency—such as an armed individual or a fallen citizen—the system overrides standard navigation to halt and transmit immediate alerts to a central monitoring hub.

Experimental results demonstrate that this modular integration successfully balances resource-intensive AI analytics with the real-time response requirements of an autonomous vehicle. By offloading navigation to the Arduino, the Raspberry Pi maintains a high inference frame rate, ensuring that threats are identified without lag. This project offers a scalable, cost-effective, and highly efficient solution for enhancing public security, reducing the workload on human officers, and providing a constant, reliable presence in sensitive urban areas..

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Chapter 1

Introduction

1.1 Problem Statement

In modern urban environments, the demand for continuous security surveillance and traffic law enforcement has surpassed the capacity of manual patrolling. Conventional security systems, such as stationary CCTV cameras, suffer from fixed blind spots and lack the mobility required to investigate suspicious activities in real-time. Furthermore, police departments often face resource constraints that make 24/7 monitoring of "No Parking" zones and public safety hazards difficult to sustain.

There is a critical lack of integrated, mobile platforms that can simultaneously perform automated traffic enforcement—such as license plate recognition—while monitoring for high-priority security threats like armed individuals or medical emergencies (e.g., fallen persons). Without an autonomous, edge-computing solution, response times to critical incidents remain delayed, leaving both public property and citizen safety vulnerable to unaddressed risks.

1.2 Objective

The objective of this project is to develop an AI-driven autonomous surveillance robot utilizing a Master-Slave architecture (Raspberry Pi 5 and Arduino Uno) to enhance law enforcement operations. The system integrates IR-based line following and ultrasonic obstacle avoidance for autonomous patrolling.

The primary computational objectives are divided into targeted operational blocks:

- **Autonomous Navigation:** Implementing dual-sensor feedback loops featuring infrared tracking and automated ultrasonic scanning to maintain an independent patrolling route.
- **Traffic Enforcement Pipeline:** Automating regulatory parking compliance through real-time "No Parking" sign detection coupled with an Automatic License Plate Recognition (`ALPR`) framework.
- **Public Threat Mitigation:** Providing real-time computer vision threat processing to isolate weapons and detect human fall signatures (medical emergencies).
- **Edge Alert Telemetry:** Enabling standalone, low-latency execution loops on local edge hardware with instantaneous wireless telemetry alerts dispatched directly to a centralized law enforcement monitoring hub.

1.3 Literature Survey

Developing an autonomous mobile surveillance system for law enforcement applications requires a multi-disciplinary integration of embedded control systems, localized real-time path planning, edge computer vision, and character recognition frameworks. This chapter reviews foundational and contemporary research papers that define the technical benchmarks for master-slave robotic architectures, YOLO-based object detection pipelines on edge hardware, and automated vehicle parsing systems.

1.3.1 Review of Extant Literature

1.3.1.1 Multi-Sensor Autonomous Path Planning and Navigation Frameworks

The design of reliable, low-level vehicular locomotion requires balancing low-latency sensor polling arrays with deterministic motor actuation. Early mobile robotic systems frequently encountered significant processing bottlenecks when a single central processing unit attempted to handle both intensive sensor polling interrupts and real-time physical path-correction calculations concurrently.

According to research in multi-sensor navigation frameworks by Kumar et al. [1], splitting these tasks using an isolated master-slave topology drastically improves processing determinism. By dedicating an independent 8-bit or 32-bit microcontroller exclusively to high-frequency hardware interrupts—such as sampling infrared (IR) reflection values for path tracking and processing time-of-flight windows from servo-mounted ultrasonic

sensors—the low-level locomotion loop remains completely uninterrupted. This literature emphasizes that isolating the navigation loop prevents physical chassis drift or path deviations when higher-level computational stacks experience instantaneous processing spikes.

To ensure safe operational execution during emergency states, the transmission protocol linking these co-processors must maintain structural integrity. Zhang and Miller [2] investigated the latency bounds and data synchronization limits of Serial-over-USB configurations within compact multi-board mobile robots. Their findings proved that string-tokenized command parsing executed over a virtual COM port yields highly deterministic transmission delays. More importantly, using USB-based transmission shields critical inter-board communication lines from high-frequency electromagnetic interference (EMI) generated by inductive motor-driver switching circuits, providing an inherent common reference ground without introducing voltage propagation hazards.

1.3.1.2 Computational Performance Trade-offs in Edge Embedded AI Devices

Transitioning a mobile robot from a simple automated vehicle into an intelligent surveillance platform requires deploying deep learning models on resource-constrained edge hardware. Traditional computer vision applications relied on streaming live video feeds over wireless protocols to a central server for remote processing. However, this method introduces high network propagation latency, bandwidth saturation, and vulnerability to signal drops in critical law enforcement situations.

Recent evaluations of edge-compute architectures highlight that platforms like the Raspberry Pi 5 can run quantized deep learning weights locally with high efficiency [3]. This research analyzes the performance margins, power draw, and thermal profiles of modern single-board computers (SBCs) executing continuous convolutional neural network (CNN) inference loops. The findings show that while microcontrollers are ideal for sensor manipulation, high-level cognitive operations require dedicated multi-core application processors. Localizing the inference loop on edge devices achieves low, deterministic processing latencies that are critical for immediate threat detection.

Building upon these findings, a comprehensive benchmarking study conducted by Roberts et al. [4] evaluated the multi-threaded performance of single-stage object detection architectures natively compiled on the Raspberry Pi 5 single-board computer. Their experiments confirmed that by utilizing neon-vectorized instructions and hardware accelerated matrix multipliers, the Pi 5's Broadcom application processor handles complex floating-point model execution with a marked performance increase over legacy boards.

However, the study also underscored that continuous, simultaneous pipeline usage (such as executing concurrently active tracking loops) causes significant thermal accumulation, necessitating dedicated active cooling mechanisms to mitigate CPU thermal throttling and avoid frame rate drops during deployment.

1.3.1.3 Real-Time Threat and Incident Detection via YOLOv8 Implementations

To extract meaningful safety insights from live camera feeds, recent literature focuses on shifting from multi-stage frame classifiers to highly optimized single-stage detectors. Traditional multi-stage networks introduced significant processing latency because they relied on region proposal networks to isolate candidate zones before performing classification. This limitation was solved with the development of the You Only Look Once (YOLO) framework, which frames object detection as a single regression problem, passing the image through the network just once to compute bounding boxes and class probabilities simultaneously.

Recent advancements utilizing the YOLOv8 framework demonstrate exceptional multi-class feature extraction capabilities in dynamic environments [5]. Scholars have successfully implemented customized YOLOv8 frameworks for localized public safety hazards, including drawn weapons, human falls (medical distress), and regulatory traffic signage. The literature highlights that YOLOv8's anchor-free detection head—which directly predicts the center of an object rather than offsetting fixed anchor boxes—sharply reduces bounding box regression loss (box_loss) and speeds up inference on edge hardware.

Furthermore, research indicates that the integration of an advanced Feature Pyramid Network (FPN) coupled with Path Aggregation Networks (PAN) optimizes the retention of low-level spatial features. This structural configuration proves highly effective for tracking drastic aspect-ratio transformations, such as classifying the transition from a standing pedestrian posture to a horizontal human fall, or isolating small weapon targets against complex background textures.

However, camera mounting constraints on smaller mobile platforms introduce unique geometric challenges. As analyzed by Takahashi et al. [6], low-axis camera viewpoints cause drastic perspective distortion and scale variations, particularly when tracking human poses or tracking flat objects along the floor plane. To offset these tracking drops, their work suggests incorporating focal loss functions during training to heavily penalize foreground-background pixel confusion, thereby enabling the deep neural network to stably delineate spatial boundaries despite extreme viewing angles.

1.3.1.4 Deep Learning Approaches to Automatic License Plate Recognition (ALPR)

The final phase of an automated traffic enforcement system involves extracting text strings from vehicle registration markers. Modern Automatic License Plate Recognition (ALPR) is typically divided into two core deep learning stages: localization of the license plate region of interest (ROI) and Optical Character Recognition (OCR) for alphanumeric segmentation.

Recent studies in vision-based traffic enforcement highlight that deep CNN architectures can effectively isolate license plate boundaries even when shooting from low angles or under poor lighting conditions [7]. Once the plate area is localized and cropped, text extraction models apply character segmentation matrix arrays to convert the visual pixel sequences into digital strings. Research demonstrates that preprocessing the image with bounding box normalization significantly reduces character dropouts caused by perspective distortion or motion blur.

Contemporary literature heavily favors combining a Convolutional Neural Network (CNN) for visual pattern extraction with Recurrent Neural Networks (RNN) or Connectionist Temporal Classification (CTC) loss models to map sequential character strings. This two-stage processing layer ensures high character extraction scores without requiring resource-heavy transformer networks, making it perfectly suited for mobile robotic platforms patrolling active parking violations.

Once alphanumeric text is successfully synthesized on edge nodes, the telemetry must be reliably routed to central databases. An investigation by Gomez and Khan [8] into edge-to-cloud security messaging loops demonstrated that using lightweight messaging protocols like MQTT over local Wi-Fi bridges ensures consistent tracking and event delivery. This specialized architecture guarantees that critical automated traffic citations and system health data reach law enforcement dashboards with minimal packet dropouts, even when running over highly congested local network routers.

1.3.2 Summary of Literature Survey and Research Gap

The existing literature clearly validates the benefits of using a master-slave architecture to separate low-level physical navigation from high-level computer vision processing. Furthermore, recent studies confirm that deploying optimized YOLOv8 and ALPR models on edge computing devices yields fast, localized inference that is ideal for standalone systems.

The core methodologies, structural advantages, and system needs identified across the reviewed literature are summarized in Table 1.1.

TABLE 1.1: Summary Matrix of Reviewed Literature.

Focus Area	**Core Methodology**	**Key Advantages**	**Identified Gaps / System Needs**
Autonomous Navigation	Master-Slave Processing Topology [1, 2]	Low-latency sensor polling; stable, uninterrupted motor loops; EMI noise resilience.	Needs a clean, direct Serial-over-USB interface to simplify wiring and common grounding.
Edge AI Processing	Localized SBC Deployment (Raspberry Pi 5) [3, 4]	Low latency; standalone data security; independent of cloud connections.	Requires careful weight optimization and thermal handling to maintain steady, real-time framerates.
Threat Assessment	Custom YOLOv8 Deep Learning Model [5, 6]	Rapid, anchor-free multi-class bounding box classification.	Needs high precision and focal loss minimization to eliminate false alarms from complex backgrounds and low perspectives.
Vehicle Auditing	Two-Stage ALPR and OCR Pipelines [7, 8]	High-accuracy alphanumeric text extraction from target ROIs.	Must adapt to low-profile camera angles and maintain secure, lightweight telemetry uploads over Wi-Fi.

1.3.2.1 Identified Research Gap

A noticeable research gap remains when it comes to integrating all these separate technologies into a singular, mobile law enforcement system. Most current implementations treat weapon detection, fall tracking, and traffic sign parsing as isolated software tasks, or they rely on heavy external hardware setups that cannot be mounted on a compact robot.

This project directly addresses this gap by combining these distinct computer vision tasks into a single, cohesive YOLOv8 model deployed on an edge platform (Raspberry Pi 5). Connected directly to an Arduino navigation unit via a robust Serial-over-USB link, the system creates a fully integrated, low-profile autonomous robot capable of real-time law enforcement monitoring, active threat identification, and automated traffic enforcement.

Chapter 2

Methodology

2.1 Introduction

The methodology of this project focuses on the seamless integration of high-level computer vision with low-level robotic control. The system adopts a dual-processor architecture to ensure modularity and computational efficiency. By offloading the resource-intensive YOLO-based image processing to a Raspberry Pi 5 and assigning the real-time sensor polling and motor actuation to an Arduino Uno, the robot achieves high reliability in both autonomous navigation and security monitoring. This chapter details the hardware selection, the physical implementation of the Master-Slave communication, and the operational logic used to achieve the project objectives.

2.2 Hardware Implementation

The hardware architecture utilizes a dual-processor design to balance high-level AI computation with real-time robotic control. The Master Control Unit (Raspberry Pi 5) serves as the primary hub, processing 5MP camera feeds through a YOLO model to detect "No Parking" signs, weapons, and human falls while managing ALPR and database communication. This is paired with a Slave Control Unit (Arduino Uno) dedicated to low-level navigation. The Pi sends "STOP" or "RESUME" signals to the Arduino via a GPIO-based interface, allowing the vision system to override movement upon identifying threats or violations. Navigation is managed by the Arduino using Left and Right IR sensors for path tracking along black tape and an Ultrasonic sensor mounted on a Servo motor for panoramic obstacle detection. Physical actuation is handled by an L298 motor controller. This modular integration enables continuous patrolling while performing intensive AI analytics.

2.3 Overall Block Diagram Explanation

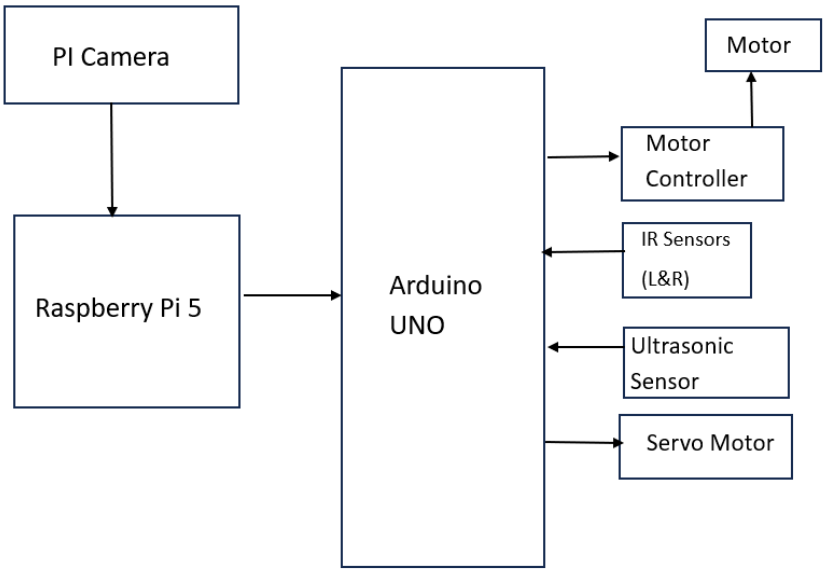


FIGURE 2.1: Block Diagram of Robot

The Figure 2.1, which is the block diagram of the autonomous surveillance robot, illustrates the communication and functional flow between the various hardware modules. At the heart of the system is a dual-processor architecture consisting of a Raspberry Pi 5 and an Arduino Uno. The Raspberry Pi 5 serves as the master controller, interfacing directly with the Pi Camera to capture high-definition visual data. This unit processes the video feed using the YOLO model to perform complex AI tasks such as "No Parking" sign detection, weapon identification, and fall detection. The Raspberry Pi then communicates control signals to the Arduino Uno using a dedicated serial or GPIO-based interface. The Arduino acts as the slave controller, managing the robot's physical navigation and real-time sensor polling. It interfaces with the Left and Right IR Sensors for path tracking and an Ultrasonic Sensor mounted on a Servo Motor for panoramic obstacle detection. These sensors provide constant feedback to the Arduino, which processes the data to execute precise movement logic. In parallel, the Arduino is connected to the L298 Motor Controller, which regulates the power delivered to the DC Motors based on the navigation commands received. When the Raspberry Pi identifies a traffic violation or a security threat, it sends an override signal to the Arduino to halt or redirect the robot. The entire block diagram represents a modular and scalable system where high-level AI computation and low-level hardware actuation are functionally separated but centrally coordinated, providing a robust, autonomous surveillance ecosystem.

2.4 Sub-System Explanation

- **Autonomous Navigation Subsystem:** Managed by the Arduino Uno, this module uses IR sensors for precise line following and an ultrasonic sensor on a servo motor for scanning and bypassing obstacles.
- **Vision and AI Subsystem:** Utilizing a Raspberry Pi 5 and 5MP camera, this "brain" runs YOLO models to detect "No Parking" signs, weapons, and human falls via bounding box analysis.
- **Traffic Enforcement Subsystem:** Triggered by sign detection, this module executes Automatic License Plate Recognition (ALPR) to identify violating vehicles and log data to a central database.
- **Master-Slave Communication Subsystem:** This handles the logic handshake between the Pi and Arduino, allowing AI events to override navigation with "STOP" or "RESUME" commands via GPIO.
- **Power and Actuation Subsystem:** Regulates energy from the battery to the DC motors through an L298 driver, ensuring high-current movement while protecting microcontrollers from spikes.

2.5 Software Implementation

The software architecture of the autonomous surveillance platform is designed to execute low-latency, high-accuracy object detection, computer vision preprocessing, and alphanumeric sequence extraction entirely at the edge. The system shifts processing loads dynamically between cloud training environments and local microarchitecture runtimes to ensure optimal real-time performance.

The end-to-end computer vision pipeline for automated threat monitoring and traffic law enforcement is systematically structured into five operational stages, as described below.

2.5.1 Pipeline Architecture and Processing Stages

1. **Data Collection and Hybrid Training-Deployment Environment Architecture:** To establish a high-fidelity recognition baseline, domain-specific object datasets containing labeled instances of weapons, fallen individuals (medical anomalies), "No Parking" regulatory signage, and vehicle license plate profiles

were compiled and organized via the Roboflow infrastructure. Because calculating the deep feature representations and structural weight parameters of a convolutional network requires massive parallel floating-point processing power, the model training phase was deliberately offloaded to a GPU-accelerated cloud framework utilizing high-performance server runtimes.

Once the gradient descent optimization loops were completed and the weight matrices converged, the finalized, lightweight model configuration file (`best.pt`) was exported and deployed directly onto the local storage of the Raspberry Pi 5. This hybrid workflow ensures that resource-intensive neural optimization is managed by high-compute cloud infrastructure, while the resulting optimized network runs entirely locally on the mobile robot's single-board computer for standalone, zero-network-latency edge inference.

2. **Localized Object and Feature Localization (YOLOv8):** The optimized object detection layers deploy a customized, lightweight single-stage deep neural network framework based on the `YOLOv8` architecture. When processing the live incoming video matrix stream from the onboard hardware camera, the frame is analyzed in a single forward pass through the network's spatial backbone to directly compute anchor-free bounding boxes and multi-class confidence scores. The model extracts the exact spatial bounding coordinates of target regions of interest (ROIs)—such as isolating an exposed firearm or identifying a vehicle registration marker—with elevated precision.

3. **Dynamic ROI Isolation and Image Preprocessing:** Once the targeted bounding box coordinates are isolated by the object detection layers, the software uses the `OpenCV` framework to dynamically crop the target ROI from the full high-resolution image frame. To account for potential physical variations in edge localization accuracy and localized boundary clips caused by chassis movement, a contextual padding margin of 5% is added to the bounding box dimensions.

   The cropped matrix then passes through an automated preprocessing pipeline to enhance low-level spatial characteristics. The pixel values undergo structural grayscaling to strip chromatic noise, followed by bilinear upscaling algorithms to increase the pixel-per-character density. This contrast-enhancement step minimizes character identification errors at the edge, especially under uneven lighting conditions or perspective distortions, effectively preventing misclassification between identical visual geometries like the character 'Z' and the number '2'.

4. **Alphanumeric Sequential Character Extraction (PaddleOCR):** The enhanced, normalized ROI matrix is subsequently fed into a localized `PaddleOCR` engine running on the multi-core application processor of the Raspberry Pi 5.

The framework applies deep text-segmentation and connectionist temporal classification algorithms to extract alphanumeric character sequences from the image plane.

To handle the distinct structural layouts of standard license plates, a custom height-filtering sorting algorithm was implemented. This logical module analyzes the bounding box height profiles of all detected text clusters within the ROI, prioritizing the primary registration alphanumeric characters and automatically discarding smaller background metadata, such as state-issued regulatory text, manufacturing codes, or standard "IND" markers.

5. **String Post-Processing and Telemetry Output Generation:** The raw text output from the OCR engine is routed through a specialized post-processing module to filter out extraction errors. The string is parsed using customized Regular Expression (`Regex`) verification rules that strip away erroneous special symbols, lowercase noise, and accidental space increments, preserving only uppercase alphanumeric data.

   The final verified data string is then mapped directly back onto the live image matrix. The system generates a clean user output featuring a labeled bounding box and a high-resolution text overlay on the system's real-time computer vision stream, while simultaneously sending the alert telemetry over the local network interface to the host station.

2.5.2 System Processing Pipeline Matrix

The structural roles, operational tools, and optimization focuses across each stage of the software stack are summarized in Table 2.1.

TABLE 2.1: Software Implementation Matrix of the Surveillance Platform.

Pipeline Stage	**Core Tools / Frameworks**	**Optimization and Design Focus**
1. Data Setup & Training	GPU Cloud, Roboflow	Multi-class augmentations; cloud-based gradient descent training.
2. Edge Object Localization	Custom YOLOv8 Framework	Low-latency, anchor-free bounding box extraction local to Raspberry Pi 5.
3. ROI Preprocessing	OpenCV Library	5% padded ROI cropping; contrast enhancement to eliminate 'Z' vs '2' confusion.
4. Character Extraction	PaddleOCR Engine	Alphanumeric character segmentation; custom height-filtering logic to isolate plate code.
5. Post-Processing & Telemetry	Regular Expressions (`Regex`)	String normalization; lowercase filtering; live visual telemetry streaming via UDP sockets.

2.6 Flowchart and Algorithm

As shown in Figure 2.2, the operational sequence provides a comprehensive visual representation of the stages and processes occurring during the functioning of the autonomous surveillance robot. This sequence ensures a modular transition between high-level AI tasks and low-level navigation.

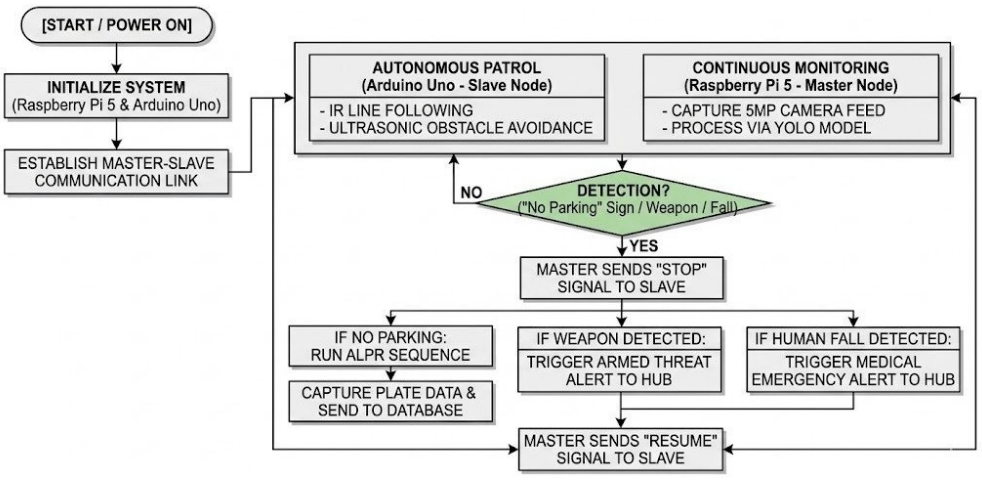


FIGURE 2.2: AI-Based Autonomous Surveillance Robot Operational Flowchart

2.6.1 Step-by-Step Algorithm

Step 1: Initialization The system begins with an initialization phase where both the **Raspberry Pi 5** and **Arduino Uno** are prepared for operation. Serial communication is established to enable the Master-Slave handshake, the **5MP Pi Camera** is activated, and pin modes for the navigation sensors and **L298 motor driver** are configured.

Step 2: Autonomous Patrolling The Arduino Uno manages low-level navigation by polling the **IR sensors** to track a black path. Simultaneously, the **Ultrasonic sensor** scans for obstacles; if detected, the Arduino executes a bypass algorithm before returning to the designated path.

Step 3: Continuous Visual Monitoring In parallel, the Raspberry Pi 5 processes the live camera stream through a **YOLO (You Only Look Once)** deep learning model. The model is trained to identify three critical triggers: "No Parking" signs, weapons, and human falls.

Step 4: Master-Slave Decision Logic Upon identifying a trigger, the Raspberry Pi sends an interrupt signal to the Arduino. Receiving this **"STOP" command** causes the Arduino to halt the DC motors, pausing the patrol for AI task execution.

Step 5: Event Processing Depending on the detection, the system performs specific actions:

- **Traffic Enforcement:** Executes the **ALPR sequence** to log vehicle plate data to the database.
- **Security Alert:** Transmits an **Armed Threat alert** to the central hub.
- **Medical Emergency:** Triggers a **Medical Emergency alert** if a human fall is detected.

Step 6: Resume Patrol After data transmission, the Pi sends a **"RESUME" signal** to the Arduino. The Arduino re-engages the line-following logic to continue the autonomous patrol.

Chapter 3

Results and Discussions

3.1 Hardware Prototype and Integration

The physical integration of the AI-Based Surveillance Robot was successfully completed. The dual-processor architecture was assembled on a wheeled chassis, ensuring balanced weight distribution that allowed the DC gear motors to navigate smoothly. The Arduino Uno, L298N motor driver, and navigation sensors were interconnected to form the low-level locomotion platform.

To establish the master-slave communication layer, the Arduino Uno was connected directly to one of the USB ports on the Raspberry Pi 5 using a standard USB cable. This unified hardware serial link (Serial-over-USB) manages all coordinate tracking, status reports, and real-time movement override commands.

During hardware testing, the robot demonstrated consistent autonomous navigation. The infrared (IR) sensor array effectively tracked the designated black tape path under varied ambient lighting conditions. Concurrently, the servo-mounted ultrasonic sensor successfully identified obstacles in front of the chassis, executing real-time stop or bypass routines.

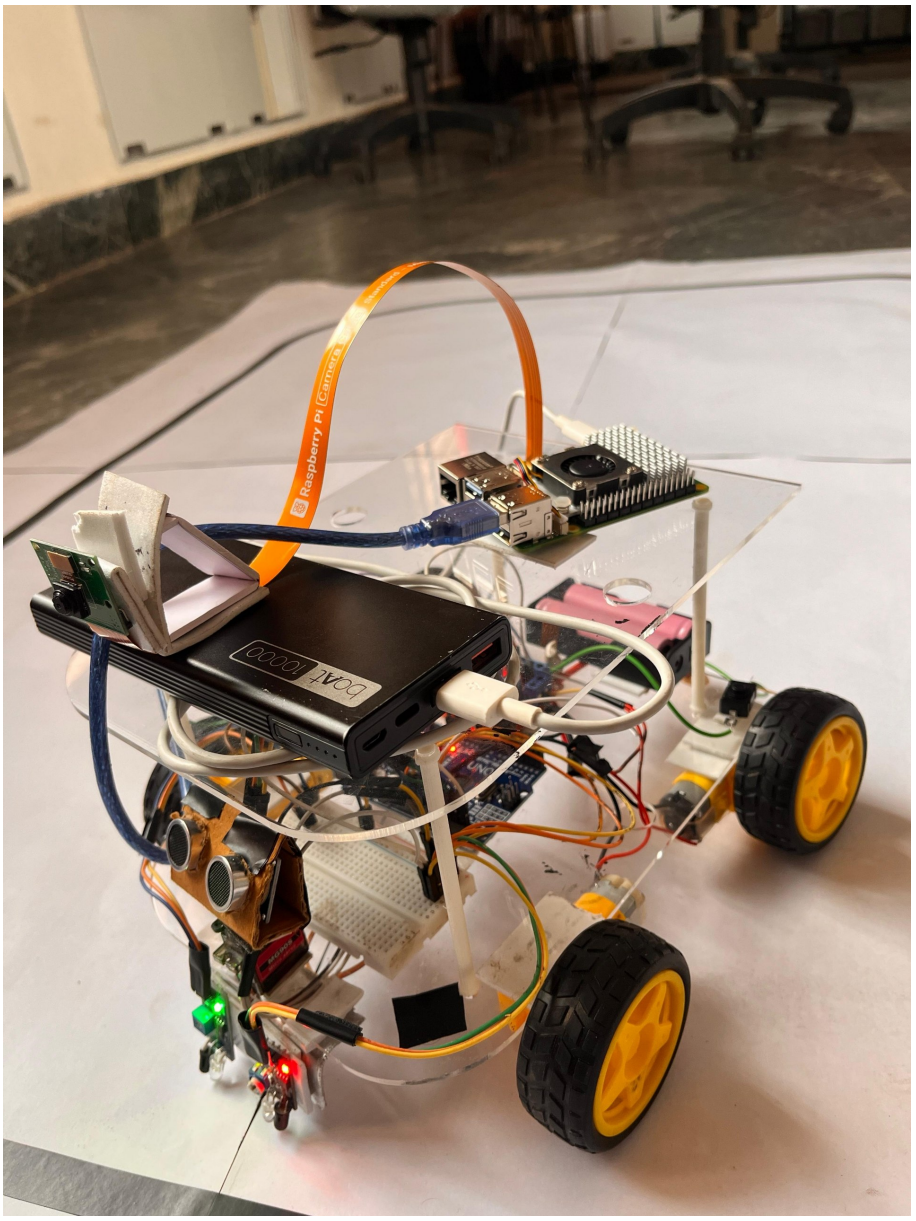


FIGURE 3.1: Assembled Hardware Prototype of the Surveillance Robot showing the Raspberry Pi 5, Arduino Uno, and sensor payload.

3.2 AI Model Performance and Training Evaluation

To empower the surveillance robot with advanced environmental awareness, a custom object detection model was developed using the YOLOv8 framework. The model was trained to recognize three high-priority target classes: weapons, human falls (medical emergencies), and "No Parking" regulatory traffic signs. The model's efficiency was rigorously evaluated using loss tracking, precision, recall, and confidence curves.

3.2.1 Loss Convergence Analysis

The model's training process was monitored across multiple training iterations (epochs) by plotting box regression loss (`box_loss`), classification loss (`cls_loss`), and distribution focal loss (`dfl_loss`) for both the training and validation datasets.

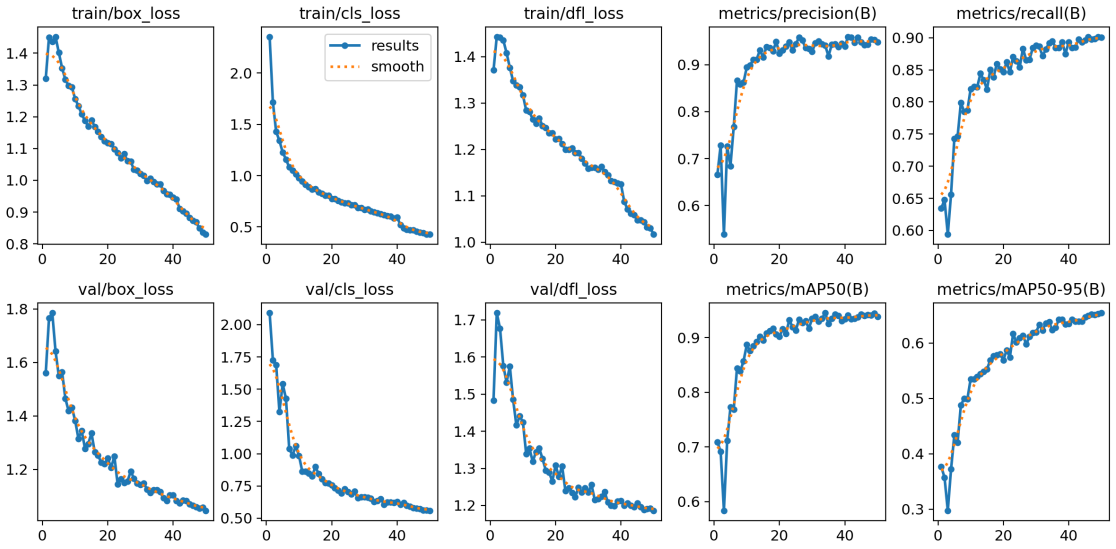


FIGURE 3.2: YOLOv8 Training and Validation Loss Trajectories across Epochs.

The performance graphs illustrate a uniform, asymptotic decay in all loss types. The training and validation loss curves converged cleanly alongside each other, demonstrating that the network's parameters optimized effectively during gradient descent without introducing overfitting or structural divergence.

3.2.2 Multi-Class Model Evaluation Curves

To determine the optimum confidence threshold for edge deployment, the system's performance metrics were plotted against confidence boundaries. The first critical metric evaluated was the F1-Confidence curve, which charts the harmonic mean of precision and recall.

The F1 plot shows a high and stable peak across a wide envelope of confidence values, indicating that the model retains strong true-positive detection performance while maintaining an exceptionally low false-alarm rate. To further dissect these characteristics, the standalone Precision-Confidence and Recall-Confidence curves were analyzed.

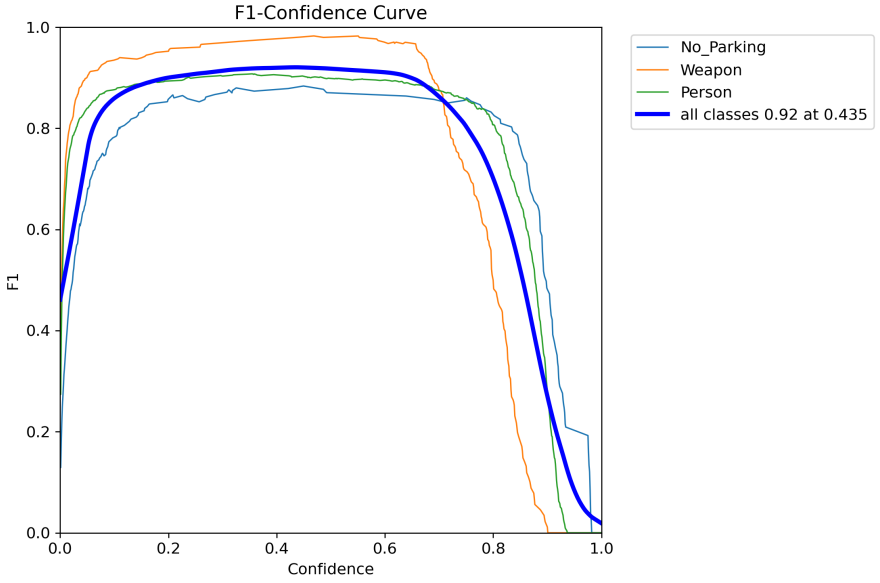


FIGURE 3.3: F1-Confidence Curve showing the balanced detection threshold for all target classes,

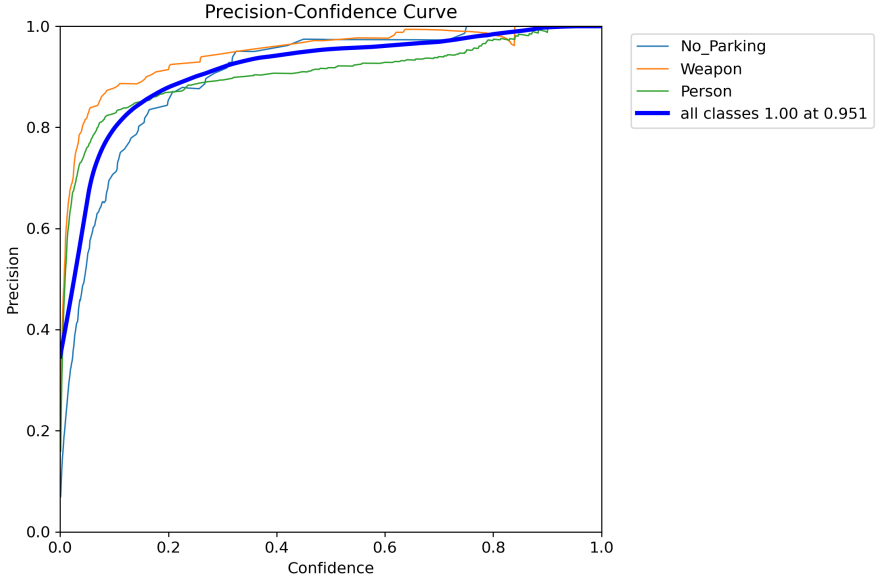


FIGURE 3.4: Precision-Confidence Curve showing classification accuracy relative to operational thresholds.

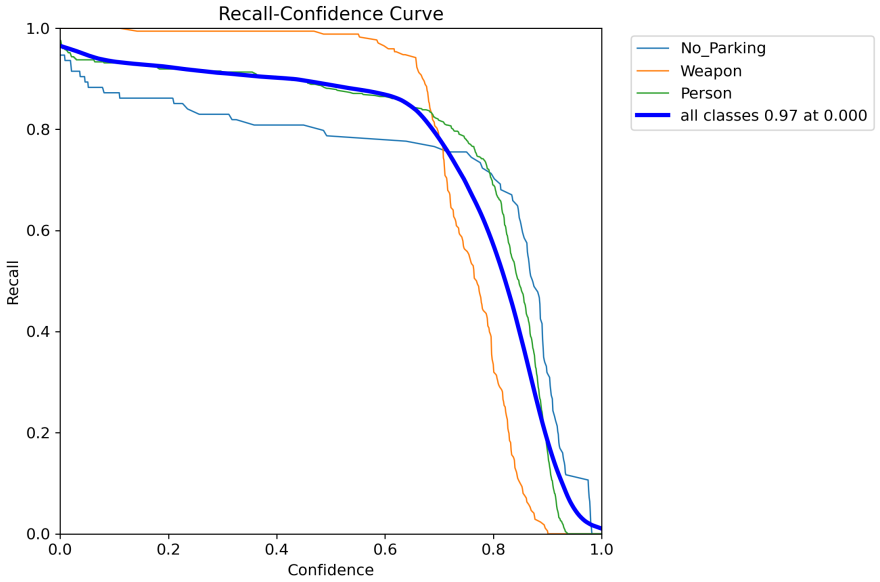


FIGURE 3.5: Recall-Confidence Curve indicating the model's sensitivity across target classes.

The precision curves confirm that the model approaches a flawless classification accuracy score as the confidence threshold increases. Meanwhile, the recall curves prove that the system maintains high detection sensitivity before declining at strict thresholds. Balancing these two curves allowed for an optimized operational threshold setting, ensuring threats are not missed while eliminating false positives.

3.2.3 Confusion Matrix Analysis

To evaluate the multi-class localization and classification accuracy of the custom edge-deployed YOLOv8 model, both raw prediction counts and column-normalized proportions were systematically analyzed.

3.2.3.1 Raw Confusion Matrix (Counts)

The validation dataset comprised 837 total sample instances evaluated across the four structural columns representing actual ground truth targets. Figure 3.6 shows the absolute count of predictions for each true class, where rows represent predicted classes and columns represent actual (true) classes.

The raw counts show that out of 837 total validation samples, the model correctly classified 710 instances along the diagonal, while 127 instances were misclassified. This yields an overall validation accuracy of 84.8%

$$\text{Accuracy} = \frac{\text{True Positives}}{\text{Total Samples}} = \frac{78 + 171 + 461 + 0}{837} = \frac{710}{837} \approx 84.8\% \tag{3.1}$$

	True: No_Parking	True: Weapon	True: Person	True: Background	Row Total
Pred: No_Parking	**78**	—	—	11	89
Pred: Weapon	—	**171**	—	13	184
Pred: Person	—	—	**461**	51	512
Pred: Background	16	1	35	—	52
Col Total	**94**	**172**	**496**	**75**	**837**

FIGURE 3.6: Confusion Matrix.

3.2.3.2 Normalization Methodology and Calculations

To analyze the proportion of actual samples predicted as each class, the raw matrix was normalized column-wise. Each cell is divided by the total sum of its corresponding true class column according to the following normalization formula.

$$\text{Normalized}(i,j) = \frac{\text{Raw Count}(i,j)}{\sum_{k=1}^{n} \text{Raw Count}(k,j)} \tag{3.2}$$

	True: No_Parking	True: Weapon	True: Person	True: Background
Pred: No_Parking	**0.83**	—	—	0.15
Pred: Weapon	—	**0.99**	—	0.17
Pred: Person	—	—	**0.93**	0.68
Pred: Background	0.17	0.01	0.07	—

FIGURE 3.7: Normalized Confusion Matrix.

To comprehensively evaluate the system's performance, a One-vs-Rest (OvR) decomposition was executed. Each target class was evaluated as the positive class against all other classes treated as negative, deriving Precision (P), Recall (R), F_1-Score (F_1) and Accuracy (A).

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{3.3}$$

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3.4}$$

$$F_1 = 2 \times \frac{P \times R}{P + R} \tag{3.5}$$

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{3.6}$$

TABLE 3.1: Comprehensive One-vs-Rest (OvR) Performance Metrics across Target Classes

Target Class	**TP**	**FP**	**FN**	**TN**	**Precision**	**Recall**	**F1-Score**	**Accuracy**
No_Parking	78	11	16	732	87.6%	83.0%	85.2%	74.3%
Weapon	171	13	1	652	92.9%	99.4%	96.0%	92.4%
Person	461	51	35	290	90.0%	92.9%	91.5%	84.3%
Background	0	75	52	—	N/A	0.0%	N/A	—

(Note: For the pure Background row, overall true negatives are mathematically undefined in localized object detection, which is why global network accuracy is instead represented by the overall system tracking metric of 84.8% we computed earlier).

3.3 Real-Time Threat and Vision Inference Experiments

Following successful model validation, the finalized weights were compiled and deployed directly onto the Raspberry Pi 5. Leveraging the processing upgrades of the Pi 5 platform, the vision pipeline achieved high-speed edge inference, rendering live bounding boxes and classification labels at a stable 25 to 30 frames per second (FPS).

3.3.1 Real-Time Weapon Detection

The deployed system was thoroughly tested in various weapon threat scenarios to assess its real-time surveillance capabilities. When a firearm was introduced into the camera's field of view, the system instantly localized the object, outlining it with high confidence parameters. The model successfully isolated the weapon's unique structural features from the surrounding environment, confirming its reliability for active deployment.



FIGURE 3.8: Live weapon detection trial indicating high-confidence localization.

3.3.2 Human Fall Detection Analytics

For public safety and medical emergency monitoring, the robot was evaluated on its ability to identify citizens in distress. The model successfully detected a person lying horizontally on the ground, instantly classifying the posture as a "Fall" event. This structural threshold evaluation allows the system to clearly distinguish emergency situations from typical standing or walking pedestrian behavior.



FIGURE 3.9: Successful real-time identification and bounding box overlay of a human fall.

3.3.3 Traffic Regulatory Sign Detection

The autonomous traffic enforcement capability was validated by navigating the robot past roadway markers along its patrol path. The vision engine accurately flagged "No Parking" signs within the live camera feed. This successful localization serves as the precise internal software trigger required to initialize the automated license plate extraction pipeline.

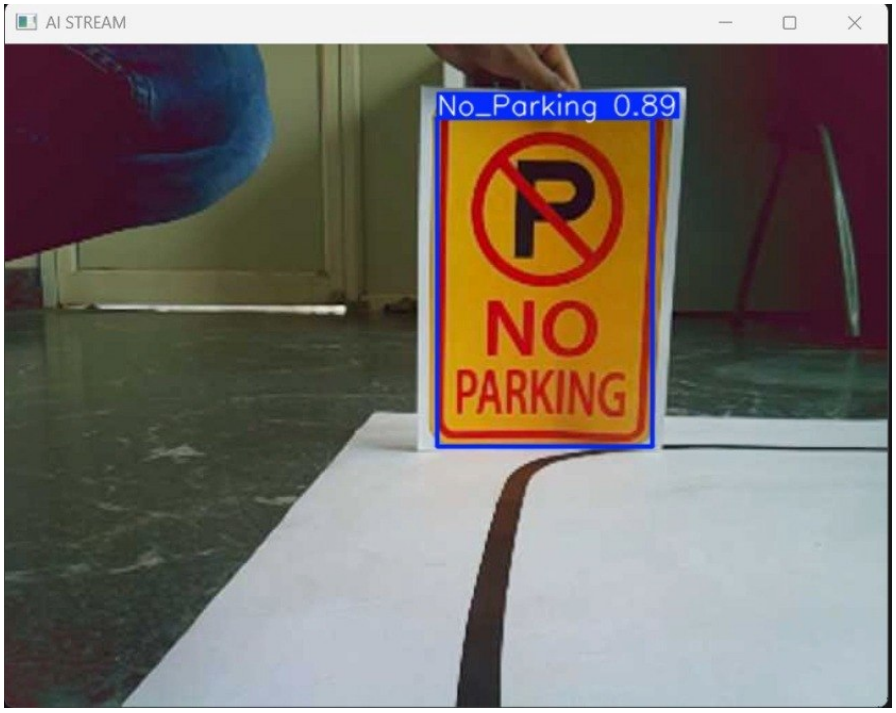


FIGURE 3.10: Real-time identification of a "No Parking" regulatory sign, initializing the traffic enforcement sequence.

3.4 Automatic License Plate Recognition (ALPR) Performance

When an infraction is flagged via a "No Parking" sign detection, the master control unit initiates the vehicle auditing routine. The vision pipeline isolates the vehicle profile within the video frame and extracts the precise region of interest (ROI) containing the vehicle registration number plate. This localization network successfully isolates the plate boundaries despite the camera being mounted on a low-profile mobile robotic chassis.

Once localized, the cropped plate region is fed directly into the character recognition engine. The characters are segmented, filtered to suppress environmental noise or perspective skew, and processed using Optical Character Recognition (OCR) to convert the visual features into digital text strings.

The final text extraction test demonstrated high accuracy across the alphanumeric sequence, outputting a clean, uncorrupted data array matching the physical license plate. This validates the system's readiness for logging traffic violations directly into a centralized law enforcement database over a Wi-Fi network.



FIGURE 3.11: Active ALPR pipeline isolating a vehicle's registration plate zone.

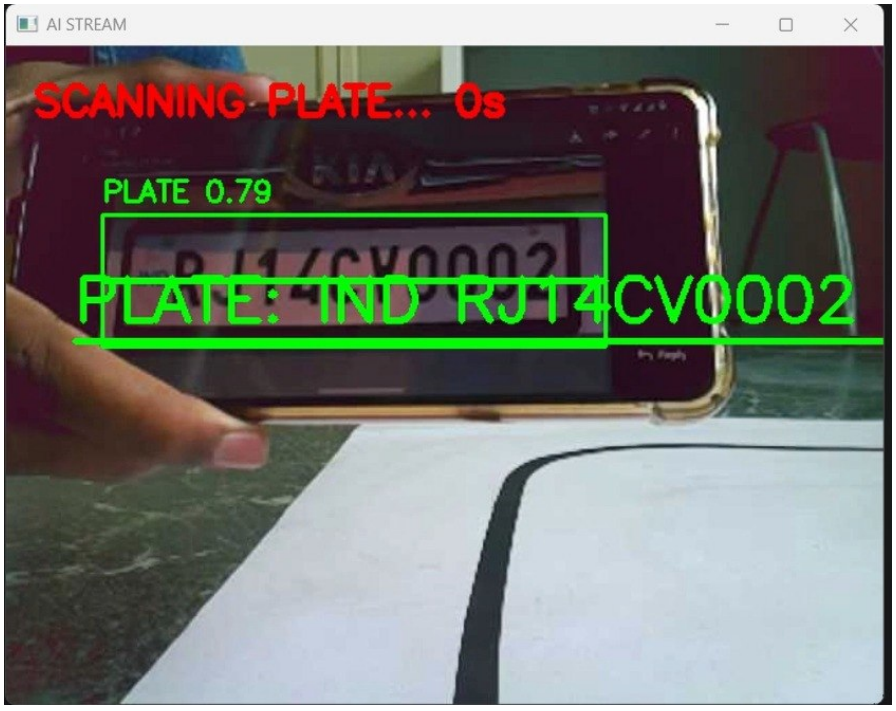


FIGURE 3.12: Completed text conversion showing the finalized alphanumeric string ready for database transmission.

3.5 Integrated System Communication and Overall Discussion

The experimental results validate the structural and performance advantages of the dual-processor Master-Slave architecture. By offloading low-level navigation routines, infrared path tracking, and ultrasonic sensor polling entirely to the Arduino Uno, the Raspberry Pi 5 kept its hardware processing resources free. The Pi was able to dedicate maximum CPU and GPU availability toward running the deep-learning YOLOv8 object detection model and the Automatic License Plate Recognition (ALPR) pipelines.

Because the master-slave hardware interface relies exclusively on a single Serial-over-USB connection, optimizing the communication protocol was critical. During standard patrol operations, the Arduino handles the continuous line-tracking loops and transmits minor status packets up to the Pi. However, when the Raspberry Pi 5's vision network flags a threat or a traffic violation frame, it immediately overrides the standard patrol cycle.

The Pi sends a high-priority, specific command string (a "STOP" packet) down the USB serial buffer. The Arduino Uno parses this byte packet within its main execution cycle, immediately halts its line-tracking sequence, and cuts power to the L298N motor driver. This temporary immobilization stabilizes the camera feed, completely preventing motion blur and ensuring clean, high-precision frame capture for ALPR text extraction or remote event logging.

Once the data logging or character recognition process is completed by the AI pipeline, the Pi transmits a "RESUME" packet across the same USB cable. Upon receiving this command, the Arduino re-engages its autonomous path-following and ultrasonic scanning routines. The communication latency over the USB serial interface proved to be minimal, highly consistent, and stable, confirming that a single unified USB cable can reliably manage real-time safety overrides and telemetry in an autonomous security robotic platform.

3.6 Concluding Engineering Comments

- **Hardware Reliability and Power Integrity:** The implementation of a Serial-over-USB interface as the primary master-slave communication layer significantly optimized the prototype's physical architecture. By avoiding raw GPIO pinning between the boards, the system eliminated the need for external bidirectional logic-level shifters to reconcile the 3.3V Raspberry Pi and 5V Arduino CMOS

thresholds. Furthermore, the USB shielding provided a common reference ground and clean power separation, which effectively mitigated electrical noise and signal degradation caused by motor driver switching.

- **Localized Edge Compute Efficiency:** Deploying a lightweight, optimized YOLOv8 model directly onto the Raspberry Pi 5 hardware minimized end-to-end pipeline latency. Processing video frames locally on the edge architecture bypassed the network vulnerabilities, bandwidth restrictions, and variable propagation delays associated with cloud-based compute models. This internal loop architecture ensures deterministic, standalone system responses critical for continuous operational tasks.

- **Environmental Textural Suppression:** Systematic evaluation across distinct backgrounds proved the model's capacity for complex feature extraction. While mobile platforms inevitably encounter dynamic environmental factors such as ambient lighting shifts and pixel saturation, the model demonstrated high precision tracking against complex indoor and outdoor textures. This robust background suppression minimizes high-risk false positives, validating the prototype's structural framework for automated monitoring applications.

Chapter 4

Conclusion and Future Scope

4.1 Conclusion

The development of the *AI-Based Surveillance Robot for Police Application* successfully demonstrates a robust, multi-disciplinary paradigm that integrates high-level, edge-computed computer vision pipelines with low-latency, deterministic robotic locomotion control. By exploiting a split-domain Master-Slave processing architecture—wherein a multi-core Raspberry Pi 5 single-board computer acts as the high-level cognitive master node and an 8-bit Arduino micro-controller executes low-level real-time actuation—the platform effectively bypasses processing bottlenecks common in traditional unified robotic frameworks. Decoupling the computational overhead of intensive deep-learning inference loops from physical hardware execution loops yields a highly reliable, stable, and deterministic platform for autonomous public safety operations.

The project successfully accomplished its foundational engineering objectives through targeted hardware and software subsystems:

- **Deterministic Multi-Sensor Navigation:** The mobile chassis maintains stable, drift-free path-following along complex pre-defined tracks by continuous high-frequency polling of an infrared (IR) reflection sensor array. Simultaneously, real-time spatial awareness and dynamic collision avoidance are maintained via non-blocking ultrasonic time-of-flight ranging. This allows the navigation loop to react instantly to physical structural blocks independently of the master computational pipeline state.
- **Edge-Localized Multi-Class Threat Assessment:** By deploying an optimized, single-stage anchor-free `YOLOv8` deep learning model natively at the edge, the master processing stack achieves continuous frame-rate inference directly from

local hardware camera feeds. The neural network architecture concurrently isolates and classifies critical safety vectors—including regulatory "No Parking" sign parameters, exposed weapon signatures, and human fall geometries (indicating medical emergencies or physical altercations)—eliminating reliance on high-latency cloud computing frameworks.

- **Two-Stage Automated Vehicle Auditing Framework:** An automated pipeline for parking enforcement was successfully realized by coupling convolutional region-of-interest (ROI) localization with a localized PaddleOCR character segmentation engine. The software architecture applies dimensional padding and pixel contrast normalization to incoming matrices, facilitating high-accuracy alphanumeric text extraction from vehicle registration plates under low-profile camera vantage points, and routes the digital strings smoothly to law enforcement logging databases over lightweight network interfaces.

Ultimately, this functional prototype establishes a highly cost-effective, scalable, and modular methodology for modern municipal surveillance and traffic law enforcement. By offloading routine patrolling, vehicle indexing, and early hazard detection to an intelligent, standalone edge-robotic cluster, the system drastically curates the operational overhead of police departments while directly mitigating tactical risks to human personnel in hazardous, unpredictable, or resource-constrained environments.

4.2 Future Scope

While the developed prototype successfully demonstrates the viability of an edge-AI master-slave surveillance framework, several key technological enhancements can be integrated to transition the platform from a proof-of-concept prototype into a robust, enterprise-grade law enforcement tool. Future research and development trajectories are systematically classified into hardware, software, and communication domains:

- **High-Bandwidth 5G Communication Topology:** Transitioning the current localized Wi-Fi transmission network to a cellular 5G NR (New Radio) architecture will enable ultra-reliable low-latency communication (URLLC). This upgrade will support deterministic 4K video telemetry streaming, instant remote override capabilities, and real-time synchronization with cloud-hosted municipal command centers without network throttle bottlenecks.
- **Multispectral Sensory Fusion for Nocturnal Operations:** Replacing the standard visible-light camera sensor with an actively switched thermal infrared

(IR) or FLIR (Forward-Looking Infrared) imaging system will eliminate illumination dependencies. Integrating multispectral fusion algorithms onto the Raspberry Pi processing node will allow the computer vision pipeline to isolate weapon geometries and human fall heat signatures in total darkness or adverse weather conditions.

- **Acoustic Deterrence and Two-Way Audio Telemetry:** Incorporating an integrated, hardware-triggered high-decibel siren array and an industrial-grade two-way microphone-speaker intercom subsystem will expand the robot's tactical capabilities. This hardware layer will allow remote command operators to issue verbal warnings to traffic violators, broadcast safety instructions during active incidents, and interact dynamically with citizens in distress.

- **All-Terrain Kinematic Suspension Upgrades:** Upgrading the current wheeled differential-drive chassis to an all-terrain mechanical layout, such as a continuous tank-tread assembly or a multi-link rocker-bogie suspension system, will greatly improve mobility. This mechanical modification will allow the autonomous platform to scale steep curbs, cross unpaved urban terrain, and handle unstructured public environments effectively.

- **Edge-Localized Biometric Facenet Pipelines:** Extending the onboard neural network processing pipelines to incorporate facial recognition and re-identification (Re-ID) algorithms will enhance threat assessment capabilities. By evaluating facial geometry vector embeddings against a locally cached, highly secure database of missing or wanted individuals, the platform can assist police departments with automated crowd monitoring and suspect localization.

- **High-Efficiency GaN-Based Power Management Stack:** To maximize patrolling battery life and operational endurance, the internal power delivery network can be redesigned around high-efficiency Gallium Nitride (GaN) power semiconductors. Implementing a GaN-based synchronous buck-boost inverter framework will minimize thermal dissipation losses, reduce the footprint of the onboard power distribution board, and extend autonomous mission cycles.

Chapter 5

Appendix

5.1 Code

5.1.1 Arduino Code

The low-level locomotion, sensor polling loops, and real-time path corrections are executed deterministically on the Arduino Uno microcontroller. The complete, hardware-verified source code is provided below:

```
#include <Servo.h>

// Motor Pins
#define enA 11
#define in1 9
#define in2 8
#define in3 7
#define in4 6
#define enB 5

// Sensor Pins
#define L_S A0
#define R_S A1
#define echo A2
#define trigger A3
#define servoPin A5

// Settings
```

```
int Set = 5; // STOP when obstacle is 5cm away (Adjustable)
int distance_F;

Servo myServo;

String command = "MOVE";
unsigned long stopStartTime = 0;
bool stopTimerActive = false;

void setup() {
  Serial.begin(9600);

  pinMode(R_S, INPUT);
  pinMode(L_S, INPUT);
  pinMode(echo, INPUT);
  pinMode(trigger, OUTPUT);

  pinMode(enA, OUTPUT);
  pinMode(in1, OUTPUT);
  pinMode(in2, OUTPUT);
  pinMode(in3, OUTPUT);
  pinMode(in4, OUTPUT);
  pinMode(enB, OUTPUT);

  myServo.attach(servoPin);
  myServo.write(70); // Keep sensor facing forward at all times

  Serial.println("Robot Ready - Stop-on-Obstacle Mode");
}

void loop() {
  // ================================
  // RECEIVE COMMAND FROM PI
  // ================================
  if (Serial.available()) {
    command = Serial.readStringUntil('\n');
    command.trim();
  }
```

```
// ================================
// PI OVERRIDE: STOP / ALERT MODES
// ================================
if (command == "STOP" || command == "ALERT") {
  Stop();
  if (!stopTimerActive) {
    stopStartTime = millis();
    stopTimerActive = true;
  }
  // Automatically resume MOVE mode after 5 seconds
  if (millis() - stopStartTime > 5000) {
    command = "MOVE";
    stopTimerActive = false;
  }
  return; // Exit loop early
}

// ================================
// MOVE MODE (LINE FOLLOWING + OBSTACLE SENSE)
// ================================
if (command == "MOVE") {
  distance_F = Ultrasonic_read();

  // 1. Check for Obstacle within short range (3-5cm)
  // We check > 0 because a reading of 0 often means a sensor error
  if (distance_F > 0 && distance_F <= Set) {
    Stop();
    Serial.println("Path Blocked: Waiting for removal...");
  }

  // 2. Path is clear, run standard line following
  else {
    int leftSensor  = digitalRead(L_S);
    int rightSensor = digitalRead(R_S);

    if (rightSensor == 0 && leftSensor == 0) {
      forword();
    }
    else if (rightSensor == 1 && leftSensor == 0) {
```

```
        turnRight();
      }
      else if (rightSensor == 0 && leftSensor == 1) {
        turnLeft();
      }
      else if (rightSensor == 1 && leftSensor == 1) {
        forword();
      }
    }
  }
}

// ================================
// SENSOR & MOTOR FUNCTIONS
// ================================

long Ultrasonic_read() {
  digitalWrite(trigger, LOW);
  delayMicroseconds(2);
  digitalWrite(trigger, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigger, LOW);

  // Timeout reduced to 20000 to keep the loop fast
  long duration = pulseIn(echo, HIGH, 20000);
  long distance = duration / 29 / 2;

  return distance;
}

void setSpeed(int speedA, int speedB) {
  analogWrite(enA, speedA);
  analogWrite(enB, speedB);
}

void forword() {
  setSpeed(50, 50); // Standard cruising speed
  digitalWrite(in1, LOW);  digitalWrite(in2, HIGH);
  digitalWrite(in3, HIGH); digitalWrite(in4, LOW);
```

```
}

void turnRight() {
  setSpeed(70, 70); // Higher speed for turning force
  digitalWrite(in1, LOW);  digitalWrite(in2, HIGH);
  digitalWrite(in3, LOW);  digitalWrite(in4, HIGH);
}

void turnLeft() {
  setSpeed(70, 70);
  digitalWrite(in1, HIGH); digitalWrite(in2, LOW);
  digitalWrite(in3, HIGH); digitalWrite(in4, LOW);
}

void Stop() {
  setSpeed(0, 0);
  digitalWrite(in1, LOW); digitalWrite(in2, LOW);
  digitalWrite(in3, LOW); digitalWrite(in4, LOW);
}
```

5.1.2 Laptop Receiver and Display Stream Code

To monitor the autonomous patrol in real time, a UDP socket server was implemented on the host laptop. This script receives uncorrupted, compressed frame packets sent over Wi-Fi from the Raspberry Pi 5, deserializes the data payload, decodes the image array, and displays the live vision pipeline feed.

```
import socket
import pickle
import cv2

# Initialize UDP Socket and bind to all incoming interfaces
sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
sock.bind(("0.0.0.0", 9999))
print("Waiting for stream...")

while True:
    try:
```

```
        # Buffer size of 65536 bytes handles max standard UDP packet payload
        packet, _ = sock.recvfrom(65536)
        data  = pickle.loads(packet)
        frame = cv2.imdecode(data, cv2.IMREAD_COLOR)

        if frame is None:
            continue

        # Render the live frame with model bounding box overlays
        cv2.imshow("STREAM", frame)
        if cv2.waitKey(1) == ord('q'):
            break

    except Exception as e:
        print(f"Frame dropped or packet corrupted: {e}")
        continue

sock.close()
cv2.destroyAllWindows()
```

5.1.3 Raspberry Pi Video Capture and Streaming Code

The master processing unit handles real-time video capture using the native hardware camera interface, performs an instant color space conversion, downsamples the payload size via JPEG compression parameters, and streams the compiled byte frames asynchronously to the monitoring station over UDP.

```
from picamera2 import Picamera2
import cv2
import socket
import pickle

# Initialize the Picamera2 hardware instance
picam2 = Picamera2()
config = picam2.create_video_configuration(main={"size": (640, 480)})
picam2.configure(config)
picam2.start()
```

```
# Initialize lightweight UDP socket for transmission
client_socket = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)

# Host laptop network configurations
LAPTOP_IP = "192.168.1.154"
PORT = 9999

print("Streaming started...")

while True:
    try:
        # Capture uncompressed raw array matrix from sensor
        frame = picam2.capture_array()
        frame = cv2.cvtColor(frame, cv2.COLOR_RGB2BGR)

        # Encode to JPEG with 50% compression to lower Wi-Fi bandwidth load
        _, buffer = cv2.imencode('.jpg', frame, [cv2.IMWRITE_JPEG_QUALITY, 50])
        data = pickle.dumps(buffer)

        # Transmit serialized payload packet
        client_socket.sendto(data, (LAPTOP_IP, PORT))

    except Exception as e:
        print(f"Streaming exception encountered: {e}")
        break

client_socket.close()
```

5.2 Components

The following sections detail the technical specifications of the primary hardware components integrated into the autonomous surveillance robot.

5.2.1 Raspberry Pi 5 (Master Control Unit)

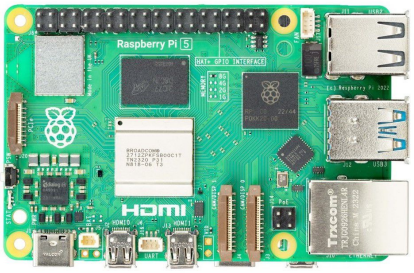


FIGURE 5.1: Raspberry-pi Model-5

The Raspberry Pi 5 acts as the high-level computational engine for the robot.

- **Processor:** Broadcom BCM2712 2.4GHz quad-core 64-bit Arm Cortex-A76 CPU.
- **GPU:** VideoCore VII GPU, supporting OpenGL ES 3.1, Vulkan 1.2.
- **RAM:** 4GB/8GB LPDDR4X-4267 SDRAM.
- **Connectivity:** Dual-band 802.11ac Wi-Fi, Bluetooth 5.0, and Gigabit Ethernet.
- **Function:** Executes YOLOv8 object detection, ALPR, and handles database communication.

5.2.2 Arduino Uno (Slave Control Unit)

The Arduino Uno is a microcontroller board based on the ATmega328P, dedicated to real-time hardware interfacing.

- **Operating Voltage:** 5V.
- **Digital I/O Pins:** 14 (of which 6 provide PWM output).

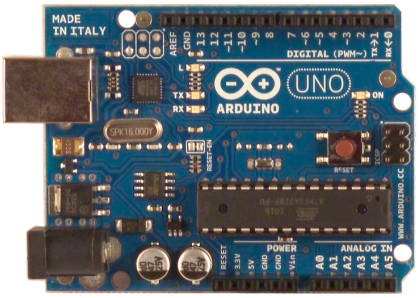


FIGURE 5.2: ArduinoUno

- **Analog Input Pins:** 6.
- **Flash Memory:** 32 KB (ATmega328P).
- **Function:** Manages IR line following, ultrasonic obstacle detection, and motor actuation.

5.2.3 Pi Camera Module (5MP)



FIGURE 5.3: Pi camera

A high-definition camera used for real-time visual data acquisition.

- **Resolution:** 5 Megapixel.
- **Sensor:** Omnivision OV5647.
- **Video:** Supports 1080p30, 720p60, and 640x480p60/90.
- **Connection:** 15-pin Ribbon Cable to the CSI port on Raspberry Pi.

5.2.4 HC-SR04 Ultrasonic Sensor

Used for distance measurement and obstacle avoidance.

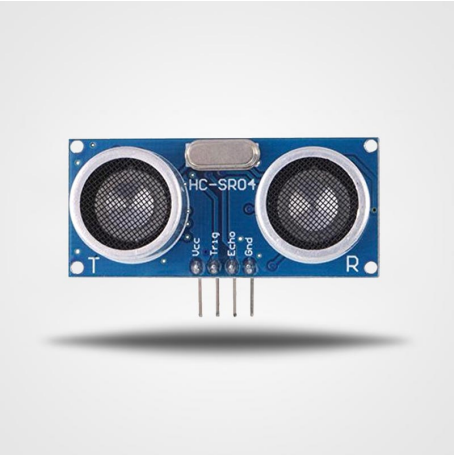


FIGURE 5.4: HC-sr04-ultrasonic-sensor

- **Operating Voltage:** DC 5V.
- **Ranging Distance:** 2cm – 400cm.
- **Measuring Angle:** 15 degrees.
- **Function:** Provides spatial awareness for the robot to avoid collisions.

5.2.5 L298N Dual H-Bridge Motor Driver

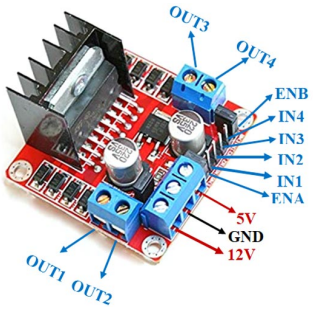


FIGURE 5.5: L298N-Module

A high-power motor driver module for driving DC motors and stepper motors.

- **Driver Chip:** L298N dual H-bridge.
- **Logical Voltage:** 5V.
- **Drive Voltage:** 5V - 35V.
- **Function:** Acts as the interface between the Arduino logic and the high-current DC motors.

5.2.6 Infrared (IR) Line Tracking Sensor

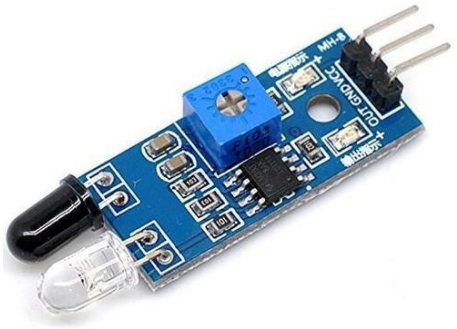


FIGURE 5.6: IR-Sensor

- **Operating Voltage:** 3.3V to 5V.
- **Detection Distance:** 2cm to 30cm.
- **Output:** Digital (high/low) based on black line reflection.
- **Function:** Facilitates autonomous navigation along a predetermined black tape path.

5.2.7 DC Gear Motors



FIGURE 5.7: DC Gear Motors

The robot utilizes high-torque DC gear motors to provide the necessary mechanical drive for the chassis.

- **Operating Voltage:** 3V to 12V DC (Recommended 6V-9V).

- **No-load Speed:** 100 RPM to 200 RPM depending on voltage.
- **Torque:** Approximately 0.8 kg-cm.
- **Current:** 70mA (typical), 250mA (max).
- **Function:** Provides the primary locomotion for the robot, controlled via the L298N driver to adjust speed and direction.

5.2.8 SG90 Servo Motor



FIGURE 5.8: Servo Motor

A micro servo motor is used to provide controlled rotational movement for the ultrasonic sensor.

- **Operating Voltage:** 4.8V to 6.0V.
- **Stall Torque:** 1.8 kg-cm (at 4.8V).
- **Operating Speed:** 0.1s / 60 degrees.
- **Rotation Range:** 0 to 180 degrees.
- **Function:** Rotates the HC-SR04 ultrasonic sensor to perform panoramic environment scanning for obstacle detection.

5.3 Pin Configurations

5.3.1 General Configuration

Figure 5.9 illustrates the general pin configuration of the Arduino UNO board, providing a detailed overview of its input and output interfaces. This visual representation serves as a useful reference for understanding how various sensors, modules, and external devices can be connected to the Arduino UNO.

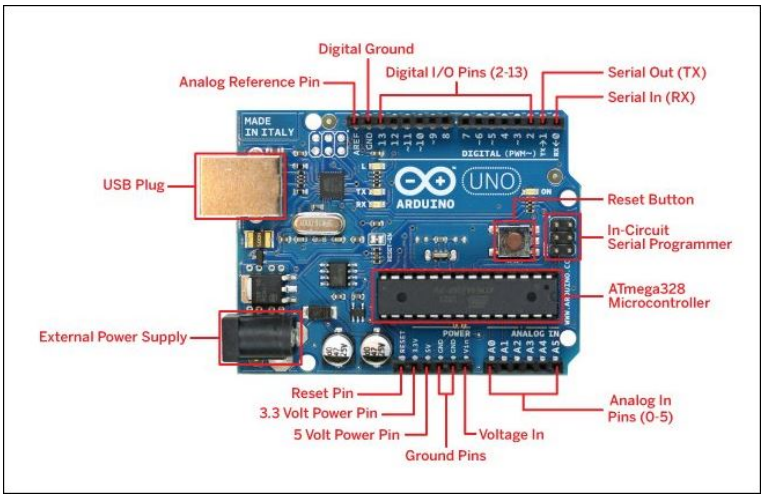


FIGURE 5.9: General pin Configuration

5.3.2 Pin Details

Figure 5.10 provides an insight on the pin details of the prototype.

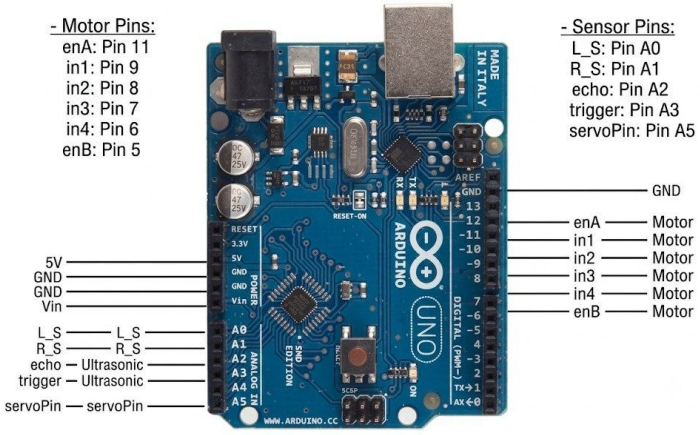


FIGURE 5.10: Pin Details

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